Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/01/2017
PI Name:	Kassemi, Mohammad Ph.D.		
Project Title:	The Zero Boil-Off Tank (ZBOT) Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE (Dec 2019): former affiliation included National Center for Space Exploration Research (NCSER), per information from J. McQuillen/GRC		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	10/05/2007	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2018 per J. McQuillen/GRC (Ed., 5/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chato, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Principal Investigator (PI): Dr. Mohammad Kassemi (Case Western Reserve University, NCSER/NASA Glenn Research Center (GRC)); Co-Investigator: Dr. David Chato (NASA GRC); Project Scientist: John Mcquillen (NASA GRC); Project Manager: Bill Sheredy (NASA GRC). NASA's projected exploration program includes a series of human and robotic expeditions to low and high Earth orbit, Moon, Mars, and possibly the asteroids and moons of other planets. Integral to all phases of these space and planetary expeditions is affordable and reliable cryogenic fluid storage for use in the propellant or life support systems. Without safe, efficient, and flexible cryogen storage, economically justified human missions may not be possible. Cryogen vaporization caused by various heat leaks into the tank from its surroundings is the main cause of storage tank self-pressurization. This has led to the development of innovative pressure control designs based on mixing of the bulk liquid with or without active cooling. Unfortunately, both tank pressurization and pressure control are governed by intricate and complicated dynamic interactions among the forced mixing, the various gravity-dependent or -independent transport mechanisms in the vapor and liquid phases, and the condensation/evaporation process at the interface. Consequently, effective implementation and optimization of a dynamic pressure control system for space applications cannot be accomplished solely by empiricism, especially, in view of serious scarcity of relevant microgravity data. Since large-scale in-space technology validation test or demonstration of the cryogenic storage tank prototype has not been possible due to both budgetary and time constraints, it was proposed that a series of small-scale targeted microgravity simulant fluid storage tank experiments can be performed aboard the International Space Station (ISS) to first understand the underlying scientific physical phenomena influencing tank pressurization/pressure control in space and then to validate and verify state-of-the-art two-phase computational fluid dynamics (CFD) models that can be used to scale-up the storage tank pressure control design for various in-orbit or on-surface applications. In this manner, the risks associated with the design of space-based storage tanks using only ground testing of the technology can be greatly reduced. In this light, the Zero-Boil-Off Tank (ZBOT) experiments have been developed as a series of small scale ISS experiments aimed at delineating important fluid flow, heat and mass transport, and phase change phenomena that affect cryogenic storage tank pressurization and pressure control in microgravity. The experiments use a simulant transparent low boiling point fluid (PnP) in a sealed transparent Dewar to study and quantify: (a) fluid flow and thermal stratification during pressurization; (b) mixing, thermal destratification, depressurization, and jet-ullage penetration during pressure control; (c) transport and kinetics effects of noncondensable pressurant gases on interfacial evaporation and condensation rates; and (d) transport, heat transfer, and phase change mass transfer interactions between evaporating and condensing sprayed droplets and ullage during active spray-bar-droplet cooling. The experiments will provide valuable microgravity empirical two-phase data associated with the above-mentioned physical phenomena through highly accurate local wall and fluid temperature and pressure measurements, nonintrusive Particle Imaging Velocimetry (PIV), and full-field phase-distribution and flow visualization. Moreover, the experiments are performed under tightly controlled and definable heat transfer boundary conditions to provide reliable high-fidelity data and precise input as required for validation & verification of state-of-the-art two-phase CFD models developed as part of this research and by other groups in the international scientific and cryogenic fluid management communities. During the project review process, the ZBOT Science Review Panel consisting of members from academia, NASA space & flight centers, and aerospace industry recommended that the broad scientific goals of the project can best be achieved through a series of consecutive experiments. Thus, the first experiment (ZBOT-1) in the series is focused on microgravity tank pressurization, thermal stratification, and destratification through mixing. The second experiment (ZBOT-2) aims at delineating the important effects of noncondensable pressurants on tank pressurization and pressure control. And the third experiment (ZBOT-3) investigates transport and phase change aspects of active tank pressure control by spray-droplet and forced-jet mixing/cooling. Implementation of the recommended strategy has resulted in a logical hierarchical series of three consecutive experiments whereupon the main tank, the fluid support unit, and the diagnostic rig are constructed during ZBOT-1 and augmented in a cost-effective step-by-step manner through implementation of additional units and diagnostics to accommodate the added complexities required by ZBOT-2 and ZBOT-3 experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The result of the ZBOT investigation will primarily and directly address the fundamental need of NASA for efficient, reliable, and enabling propellant and life support technologies for long-duration planetary exploration. However, the fundamental two-phase data and scientific understanding provided by the experiment and the associated theoretical model development will affect a diverse set of ground-based two-phase technologies such as condensing heat exchangers, phase separators, water-handling operations in fuel cells, and storage of cryogenic and non cryogenic consumable fluids.
	ZBOT 2016-2017 Task Progress Excellent hardware development, testing and model validation progress was made this year in the ZBOT project on five fronts: 1. Hardware development & testing 2. Ground-based experimentation 3. CFD model development, Verification & Validation 4. Dissemination of results through archival publication and conference presentations 5. Linkage with NASA Space Technology Mission Directorate (STMD). 1. Hardware, Experimental Procedures, and Project Development: Several problems surfaced during off-gas and other testing of the hardware such as leak issues within the Fluid Support Unit (FSU) relief valves, a solenoid valve, and the membrane contactor and disconnected RTDs. The problems were resolved. The system was reassembled and successfully leak tested. Off gas testing was repeated and successfully passed. There were also issues with the PIV particle injectors that were also resolved in a satisfactory manner. The Mission

Task Progress:	Sequence Testing (MST) was performed and engineering requirement verification activities were completed. The System Acceptance Reviews (SAR-1 and SAR-2) were conducted in January 2017 and also successfully passed. Detailed discussions were held to streamline flight testing procedures for fluid preconditioning, pretest tank preparation, and initiation and termination of test runs. PIV, white light, instrumentation data taking, sequencing, storage, and download procedures were also reexamined and streamlined. Work on a detailed flowchart and Matlab code to process and present the downloaded flight data was also initiated and is continuing. ZBOT flight system was packaged and delivered to Cape Canaveral and loaded on OA-7 Rocket. It flew to the ISS on April 18, 2017. The project team is preparing for an Operational Readiness Review (ORR) in June, 2017 and microgravity operations are scheduled to begin in the timeframe of July – Sept 2017. 2. Ground-based Experimentation & Testing The Thermal Science Test (TST) and the Mission Sequence Test (MST) were both successfully performed this year by the ZIN engineering team. Both tests were performed in the flight hardware but only the MST used the flight software to perform the entire test. Self-pressurization tests were conducted in two modes: Vacuum Jacket and strip heating in attempt to simulate heat leaks from the environment or the support structure. In order to have consistent data conducive for model validation, all the self-pressurization tests are performed from the same well-established and quantifiable uniform temperature initial condition established by intervals of constant temperature mixing and hold during the pre-test tank preparation. The pressure control studies are performed either from an elevated uniform temperature condition or from thermally stratified conditions following a self-pressurization run. Liquid is drawn from the tank, passed through the heat exchanger in the FSU, and then injected into the tank at a given flow rate either at the average tank fluid temperature for mixing only and ullage penetration studies or at a prescribed subcooled level for active cooling pressure control tests. Jet flow rates varied from 2-25 cm/s. The TST constituted 8 tests covering self-pressurization runs by both Strip Heater and Vacuum Jacket heating and jet mixing runs with different level of jet temperature subcooling at the 80% and 90% fill levels. The data from MST was used to test the Matlab post processing code developed in-house. Experimental results from both TST and MST are currently being used to validate the CFD models with 1g pressurization and pressure control data. 3. CFD model development, Verification & Validation Hand-in-hand with the experiments, development of comprehensive 2D and 3D conjugated two-phase Computational Fluid Dynamics (CFD) models for simulation and prediction of storage tank pressurization and pressure control is substantially progressed during this year. In order to delineate the intricate transport and phase change phenomena underlying tank pressurization and pressure control, two classes of two-phase CFD models are developed. The 2D and 3D versions of both models are based on the detailed hardware geometry constructed by engineering CAD packages. The first two phase CFD model assumes a stationary sharp interface between the ullage and the liquid and solves the momentum and energy equations in both phases. The species equation is also solved in the ullage region to account for the transport of residual noncondensables gas that coexists in the vapor region. The interfacial energy and force balances and the Stefan wind relationship for evaporation/condensation process are rigorously formulated and applied at the sharp interface boundary. The kinetically based Schrage equation is used to compute the evaporating/condensing mass flux at the interface and the dependence of surface tension on temperature is also included to account for Marangoni convection for those situations that the interfacial temperature varies spatially with the vapor partial pressure or when it departs from its saturation value. In the second model, the interface is allowed to move according to the forces imposed upon it. Here, a volume of fluid (VOF) scheme is used to capture the movement of the interface based on an evolution equation for the gas-liquid volume fraction. Essentially, the same set of equations as in the previous model are solved in the two phases except for the fact that in the VOF implementation, the interfacial balances are satisfied on a volumetric basis using source terms in the interfacial region as marked/captured by the volume fraction. Again, the Schrage equation is used to evaluate the evaporation/condensation mass flux and the Brackbill continuum source force model is used to account for the surface tension forces driving the Marangoni convection. At this stage, the CFD models are partially verified and validated with ZBOT ground-based data and existing large-scale 1G LH2 experiments performed by NASA. The models will be further validated in the upcoming year by the microgravity data provided by the ZBOT experiments for Space applications. 4. Archival Publication & Conference Presentations In 2016-2017 the PI and Science Team presented the results of ZBOT ground-based activity in 2 archival journal publications and 6 conference presentation/papers. 5. Linkage with NASA Space Technology Mission Directorate (STMD) & NASA Exploration Program In 2017-2018, ZBOT microgravity data is going to play a large role in validation of the CFD models and codes developed under the Development and Validation of Analysis Tools (DVAT) task of the STMD Evolvable Cryogenics (eCryo) project. As it is currently planned, three different classes of models including both multi-node system level engineering and first principal two-phase CFD models developed by the PI and other groups at NASA Glenn Research Center and NASA Marshall Space Flight center will be validated against the ZBOT data. ZBOT has also been subject of two presentations and working group discussions by the PI with the Japanese (JAXA) and German (DLR) space agencies with potential for future international collaboration and model validation efforts.
Bibliography Type:	Description: (Last Updated: 07/29/2025)

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